

Brac University

Department of Electrical & Electronic Engineering

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Course Number : EEE103L

Course Title : Computer Programming Laboratory

Section :06

Group No : 03



Lab Report

Experiment no:

Name of the experiment:

Arrays and string in C language.

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Group contribution:

Writing : Souvik Barman Ratul

Coding : Shakila Hoque Nanjiba

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But each member of group 3 contributed to each segment by working together collaboratively.

Title and Objective:

- Title: Arrays and Strings in C Language
- Objective: To learn how arrays and strings work in C and how to use them for storing, accessing, and manipulating data.

Theoretical background:

- Array: A collection of data of the same type stored in continuous memory. Each element is accessed by index (starts at 0).
Example: `int num[5] = {1,2,3,4,5};`
- String: A set of characters stored as an array and ending with '\0'.
Example: `char name[] = "Hello";`
- Arrays are useful for handling lists of numbers.
- Strings are useful for working with text and words.

Lab report assignments

a. Write a program in C using an array which finds the largest and smallest elements in a group of numbers.

Answer:

```
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter number of elements: ");
    scanf("%d", &n);

    int arr[n];

    printf("Enter %d numbers:\n", n);
    for(i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
    }

    int largest = arr[0], smallest = arr[0];
    for(i = 1; i < n; i++) {
        if(arr[i] > largest)
            largest = arr[i];
        if(arr[i] < smallest)
            smallest = arr[i];
    }

    printf("Largest = %d\n", largest);
    printf("Smallest = %d\n", smallest);
    return 0;
}
```

Output

```
Enter number of elements: 2
Enter 2 numbers:
3,2
Largest = 3
Smallest = 0
```

=== Code Execution Successful ===

b. Write a program in C to augment one string with another string. For example: String 'a' contains "foot", string 'b' contains "ball", now string 'c' should contain "football".

Answer:

```
#include <stdio.h>

#include <string.h>

int main() {

    char a[100];

    char b[100];

    char c[200];

    printf("Enter first string: ");

    scanf("%s", a);

    printf("Enter second string: ");

    scanf("%s", b);

    strcpy(c, a);

    strcat(c, b);

    printf("Final string: %s\n", c);

    return 0;

}
```

Output

```
Enter first string: foot
Enter second string: ball
Final string: football

=== Code Execution Successful ===
```

c. Write a program to print a string in reverse order.

Answer:

```
#include <stdio.h>

#include <string.h>

int main() {

char str[100];

int i;

printf("Enter a string: ");

scanf("%s", str);

int len = strlen(str);

printf("Reversed string: ");

for(i = len - 1; i >= 0; i--)

{

printf("%c", str[i]);

}

printf("\n");

return 0;

}
```

Output

```
Enter a string: 1,5,3
Reversed string: 3,5,1
```

```
=== Code Execution Successful ===
```

d. A palindrome is a word, number or other sequence of characters which reads the same backward as forward, such as 'madam'. Write a C program which can determine if a word is palindrome or not.

Answer:

```
#include <stdio.h>

#include <string.h>

int main() {
    char str[100];
    int i;
    int len;
    int flag = 1;
    printf("Enter a word: ");
    scanf("%s", str);
    len = strlen(str);
    for(i = 0; i < len / 2; i++)
    {
        if(str[i] != str[len - i - 1])
        {
            flag = 0;
            break;
        }
    }
    if(flag)
        printf("Palindrome\n");
    else
        printf("Not a palindrome\n");
    return 0;
}
```

Output

Enter a word: madam

Palindrome

=== Code Execution Successful ===

Discussion:

In this experiment, we practiced how arrays and strings are used in C programs. With arrays, we learned to store and process multiple values easily, such as finding the largest and smallest numbers. With strings, we explored operations like combining two strings, reversing a string, and checking if a word is a palindrome. These exercises helped us understand how arrays handle numeric data and how strings handle text. Overall, the experiment showed the importance of arrays and strings in solving real problems in programming.